

11

NUCLEAR PHYSICS

CHAPTER OUTLINE

- | | |
|--|------------------------------------|
| 11.1 The Nucleus | 11.5 Nuclear Binding Energy |
| 11.2 Radioactivity | 11.6 Nuclear Fission |
| 11.3 Half-Life | 11.7 Nuclear Fusion |
| 11.4 Artificial Nuclear Reactions | |

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11.1 The Nucleus

- ❑ The *nucleus* is unimaginably tiny yet incredibly dense and occupies the very center of the atom.
- ❑ The nucleus contains two kinds of particles: *protons* and *neutrons*.
- ❑ These particles have nearly the same mass, which is about 1,840 times that of an electron.
- ❑ The masses are extremely small, so it becomes convenient to introduce an appropriate unit of mass, *the atomic mass unit, u*, equal to:

$$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$$

11.1 The Nucleus

- Using this measure, the masses of a proton and a neutron are each about 1 u.
- The proton carries a positive charge, and the neutron is uncharged.
- The number of protons in the nucleus is the atomic number Z of the atom.

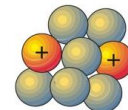
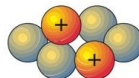
Particle	Mass	Charge
Electron	$9.109 \times 10^{-31} \text{ kg} = 0.00055 \text{ u}$	$-1.602 \times 10^{-19} \text{ C}$
Proton	$1.67262 \times 10^{-27} \text{ kg} = 1.00730 \text{ u}$	$1.602 \times 10^{-19} \text{ C}$
Neutron	$1.67493 \times 10^{-27} \text{ kg} = 1.00869 \text{ u}$	0

11.1 The Nucleus

- The uncharged **neutrons** have **influence on the atom's mass**.
- In fact, the number of neutrons in the nuclei of a particular element can vary.
- For example, most helium atoms have two neutrons in the nucleus, but some have one, three, four, or even six. Each is **still a helium atom; the only difference is their mass**.
- This allows us to associate another number with each nucleus, ***the neutron number***.

 Proton

 Neutron



11.1 The Nucleus

DEFINITION

Neutron Number: The number of neutrons contained in a nucleus.

The symbol N is used to represent *the neutron number*. For all helium atoms, $Z = 2$, but N can be 1, 2, 3, 4, or 6.

11.1 The Nucleus

- ❑ The *mass of an atom* is determined primarily by how many protons and neutrons there are in the nucleus.
- ❑ The electrons are so light that they contribute only a negligible amount to the total mass. Therefore, it is useful to define yet a third number for each nucleus.

DEFINITION

Mass Number: The total number of protons and neutrons in a nucleus.

The *mass number* is represented by the letter *A*.

$$A = Z + N$$

11.1 The Nucleus

- ❑ The *mass number indicates what the mass of a nucleus* is, just as *the atomic number indicates the amount of electric charge in the nucleus*.
- ❑ *Protons and neutrons are collectively referred to as nucleons*.
- ❑ The *mass number* is just the total number of nucleons in the nucleus.
- ❑ For example, the possible mass numbers for helium are $A = 3, 4, 5, 6, \text{ or } 8$. Each different possible “type” of helium is called an *isotope*.

DEFINITION

Isotopes: Isotopes of a given element have the same number of protons in the nucleus but different numbers of neutrons.

11.1 The Nucleus

- ❑ The different **isotopes** of an element have essentially **the same atomic properties**: these are determined by the atomic number Z .
- ❑ The different **isotopes** of an element often do have vastly **different nuclear properties**.
- ❑ Most of the 118 different elements have several isotopes. Some have only a few (hydrogen has 3), and others have more than 20. More than 3,100 different isotopes have been identified. Of these, only about 340 occur naturally. The rest are produced in certain nuclear processes.

PERIODIC TABLE OF THE ELEMENTS

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54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72
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73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91
As	Se	Br	Kr	Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Pb
92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	Nb	Mo	Tc	Ru	Rh	Pd	Ag
111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129
La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu	Hf	Ta	W	Re
130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148
Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	Nb	Mo	Tc	Ru	Rh	Pd
149	150	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167
Fr	Ra	Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	Nb	Mo	Tc	Ru
168	169	170	171	172	173	174	175	176	177	178	179	180	181	182	183	184	185	186
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225	226	227	228	229	230	231	232	233	234	235	236	237	238	239	240	241	242	243
Fr	Ra	Ac	Th															

Notes:

- (1) Tan-solid
Red-gas
Blue-liquid
Green-artificially prepared

- (2) Values for gaseous elements are for liquids at the boiling point.

- (3) Numbers in parentheses are uncertain.

11.1 The Nucleus

- ❑ Each different **isotope** of an element is *designated by its mass number*.
- ❑ For example, the most common isotope of helium has $A = 4$ ($Z = 2$ and $N = 2$) and is called helium-4.
- ❑ The three isotopes of carbon in your body are carbon-12, carbon-13, and carbon-14.
- ❑ Two of the isotopes of hydrogen have been given special names: Hydrogen-2 is called deuterium, and hydrogen-3 is called tritium.
- ❑ We use a **special notation to represent each isotope**. It consists of the atom's chemical symbol with a subscript and a superscript on the left side. **The subscript is the atom's atomic number Z** , and the **superscript is the atom's mass number**. Some examples are shown here:

Helium-4: ${}^4_2\text{He}$

Carbon-14: ${}^{14}_6\text{C}$

Carbon-12: ${}^{12}_6\text{C}$

Uranium-235: ${}^{235}_{92}\text{U}$

11.1 The Nucleus

Helium-4: ${}^4_2\text{He}$

Carbon-14: ${}^{14}_6\text{C}$

Carbon-12: ${}^{12}_6\text{C}$

Uranium-235: ${}^{235}_{92}\text{U}$

- ❑ The two numbers of A and Z indicate the relative mass and charge that the nucleus possesses.
- ❑ The **neutron number N** can be *found by subtracting the lower number (Z) from the upper number (A)*.
- ❑ For example, each uranium-235 nucleus contains 143 neutrons ($235 - 92 = 143$).
- ❑ Note that both the chemical symbol and the subscript indicate what the element is, so they must always agree. Regardless of what the superscript is, if the subscript is 6, the element is carbon and the symbol must be C.

11.1 The Nucleus

- ❑ Freezing, boiling, burning, crushing, and other chemical and physical processes *do not affect the nuclei of atoms*.
 - *These processes are influenced by the forces between different atoms, forces that involve only the outer electrons.*

- ❑ But nuclei are not indestructible: a number of *nuclear processes do affect them*. Nuclei can lose or gain neutrons and protons, absorb or emit gamma rays, split into smaller nuclei, or combine with other nuclei to form larger ones.
 - These processes are called *nuclear reactions*. Some occur around us naturally, but most are produced artificially in laboratories.

11.1 The Nucleus

This notation can be extended to represent individual particles as well. The designations for neutrons, protons, and electrons are shown here:

Neutron: ${}^1_0\text{n}$

Proton: ${}^1_1\text{p}$

Electron: ${}^0_{-1}\text{e}$

- For *the electron*, *0 is used* for the mass number because its mass is so small. The *-1 indicates* that it has the same size *charge* as a single proton, except it is *negative*.

11.1 The Nucleus

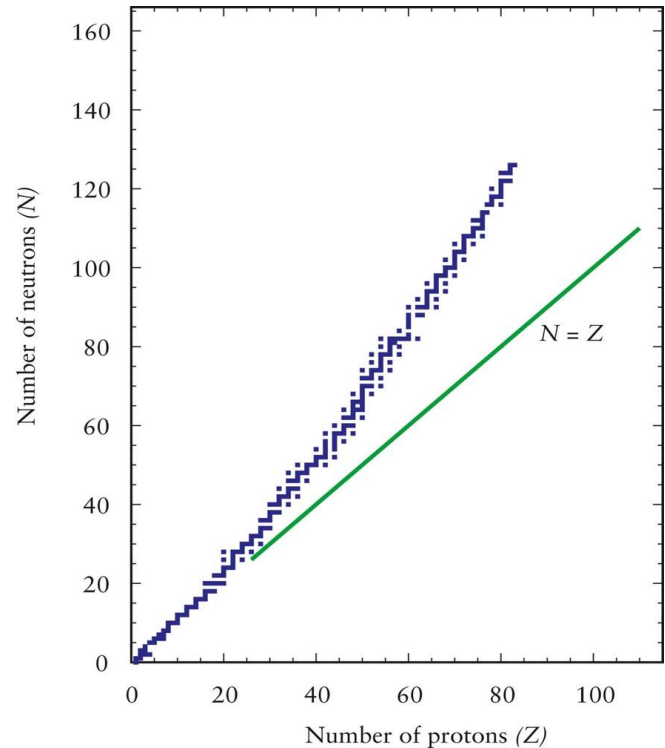
- ❑ Perhaps you've wondered how protons can be bound together in a nucleus. After all, like charges do repel each other.
- ❑ There is indeed a large electrostatic force acting to push the protons apart, but *another force many times stronger acts to hold them together inside a nucleus*. This is called *the strong nuclear force*.
- ❑ The *strong nuclear force* is *an attractive force that acts between nucleons irrespective of their charge*: every proton and neutron in a nucleus exerts an attractive force on every other proton and neutron.
- ❑ It is much stronger than the other forces, but *it has an extremely short range*: the attractive nuclear force between particles in the nucleus begins to weaken appreciably when the particles are more than about 3×10^{-15} meters apart.

11.1 The Nucleus

- ❑ Because the strong nuclear force has an extremely short range, this puts an upper limit on the size that a nucleus can have and still be stable.
- ❑ In a large nucleus, protons on opposite sides are far enough apart that they are near the limit of the effective range of the strong attractive nuclear force. In these instances, the repulsive electric force between the protons becomes important.
- ❑ *No known stable isotope has an atomic number larger than 83.*

11.1 The Nucleus

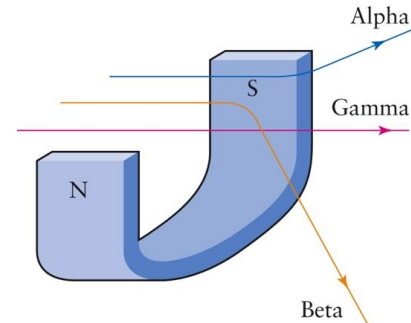
- ❑ *The effectiveness of the nuclear force at holding a nucleus together also depends on the relative numbers of neutrons and protons.*
- ❑ If there are too many or too few neutrons compared to the number of protons, the nucleus will not be stable.
- ❑ In the plot each small blue square represents one stable isotope.
- ❑ The green line indicates the trend for nuclei that would have the same number of protons and neutrons. The points curve away from this line, indicating that larger stable nuclei have successively more neutrons than protons.



11.2 Radioactivity

Radioactivity, also called **radioactive decay**, is a process wherein an unstable nucleus emits particles or EM radiation (or both).

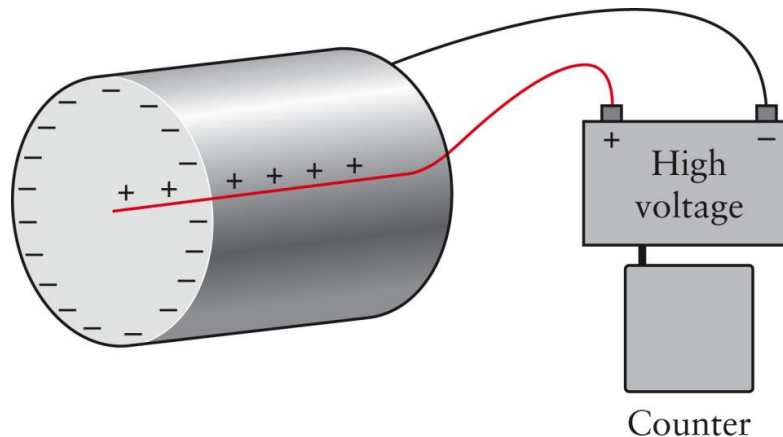
- Isotopes with unstable nuclei are called **radioisotopes**. The majority of all isotopes are radioactive.



- There are three different types of nuclear radiation named: **alpha**, **beta**, and **gamma** radiation.
- Alpha rays** and **beta rays** are *high-speed charged particles*. They can be *deflected* with magnetic and electric fields.
- Gamma rays** are extremely high-frequency *electromagnetic waves*, unaffected by the magnetic and electric fields.

11.2 Radioactivity

- ❑ To detect radiation, the most common device is the *Geiger counter*.
- ❑ It exploits the fact that nuclear radiation is ionizing radiation.
- ❑ Most Geiger counters emit an audible click each time alpha, beta, or gamma ray is detected and also keep track of their count rate.



11.2 Alpha Decay

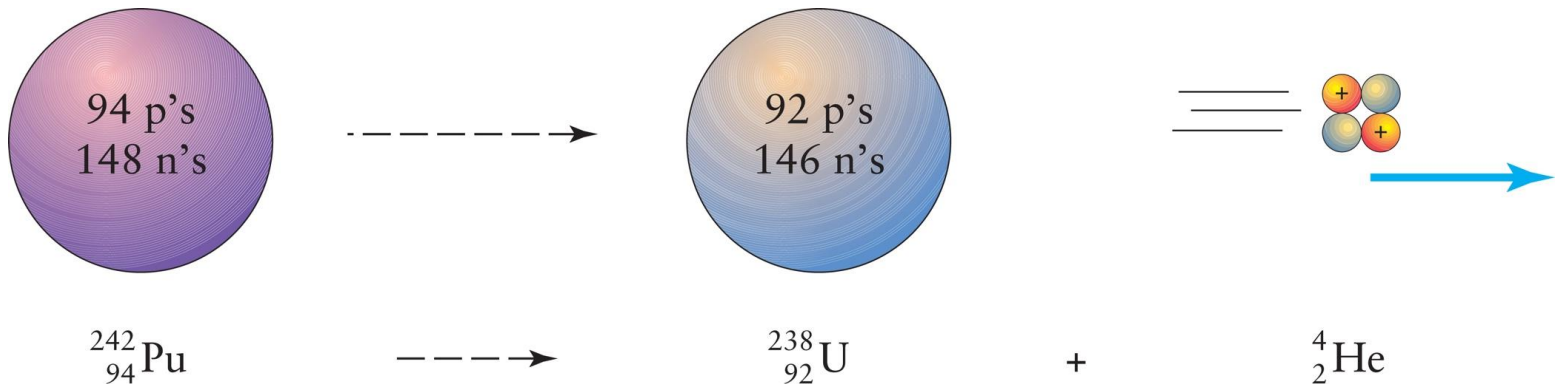
An *alpha particle* is really four particles tightly bound together: two protons and two neutrons. It is identical to a nucleus of *helium-4*. For this reason, an alpha particle can be represented as:

alpha particle: α or ${}^2_4\text{He}$

- A nucleus that undergoes alpha decay does not contain a helium-4 nucleus. This just happens to be a particularly stable combination of nuclear particles that can be ejected as a group from the nucleus.

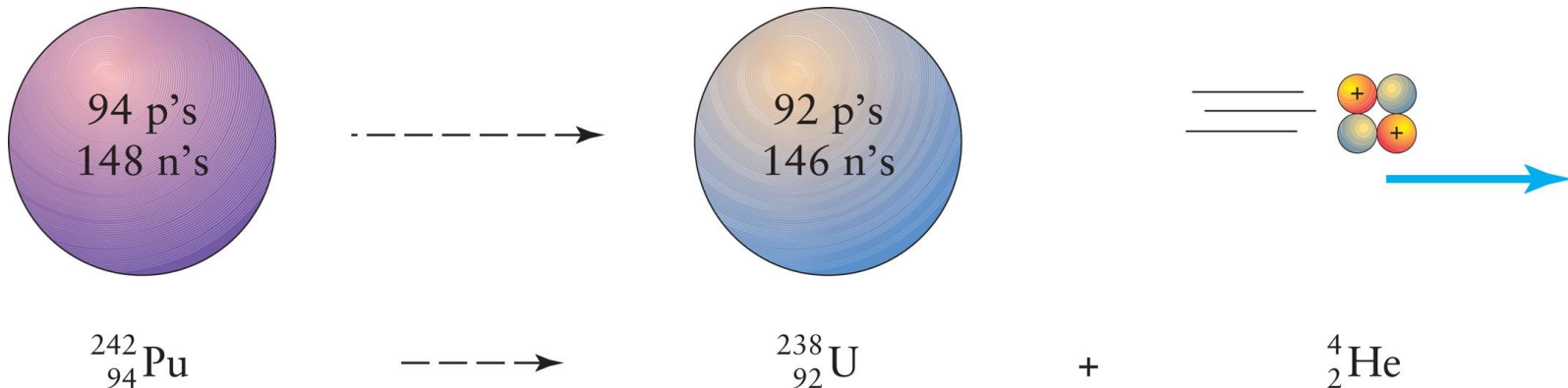
11.2 Alpha Decay

- ❑ The emission of an alpha particle *reduces both the atomic number Z and the neutron number N by 2*. The *mass number A is reduced by 4*.
- ❑ The figure is a diagram of the alpha decay of a plutonium-242 nucleus. The atomic number of the nucleus decreases from 94 to 92: the nucleus is transformed from plutonium to uranium. The original plutonium-242 nucleus is called the **parent**, and the resulting uranium-238 nucleus is called the **daughter**.



11.2 Alpha Decay

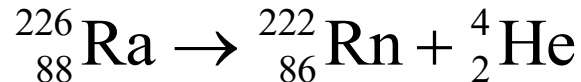
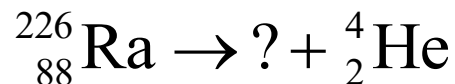
- ❑ Because alpha decay causes such a drastic change in the mass of the nucleus, it generally occurs only in radioisotopes with high atomic numbers.
- ❑ Alpha particles are quickly absorbed when they enter matter: even something as thin as a sheet of paper can stop a significant fraction of them.



Example 11.1

The isotope radium-226 undergoes alpha decay. Write the reaction equation and determine the identity of the daughter nucleus.

From the periodic table of the elements, we find that the atomic number of radium is 88 and its chemical symbol is Ra. So the reaction will appear as shown here.

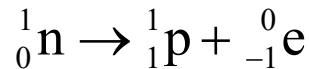


11.2 Beta Decay

A **beta particle** is simply an electron ejected from a nucleus. This means that a beta particle has the same symbol as an electron.

Beta particle: β or ${}_{-1}^0\text{e}$

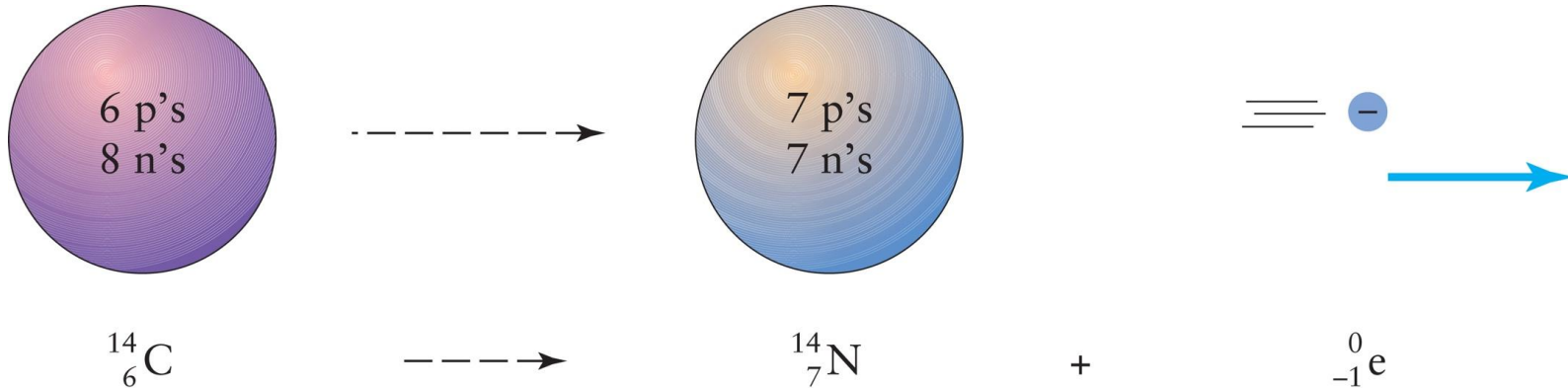
neutron $\rightarrow$ proton + electron



- But there aren't any electrons in a nucleus! During beta decay, *one of the neutrons is spontaneously converted into an electron (the beta particle) and a proton.*
- The electron is ejected with very high speed, and the proton remains in the nucleus.

11.2 Beta Decay

A nucleus that undergoes **beta** decay **loses one neutron** and **gains one proton**. The figure shows the beta decay of a carbon-14 nucleus. Note that the mass number A of the nucleus is unchanged, but now the atomic number is reduced by one.

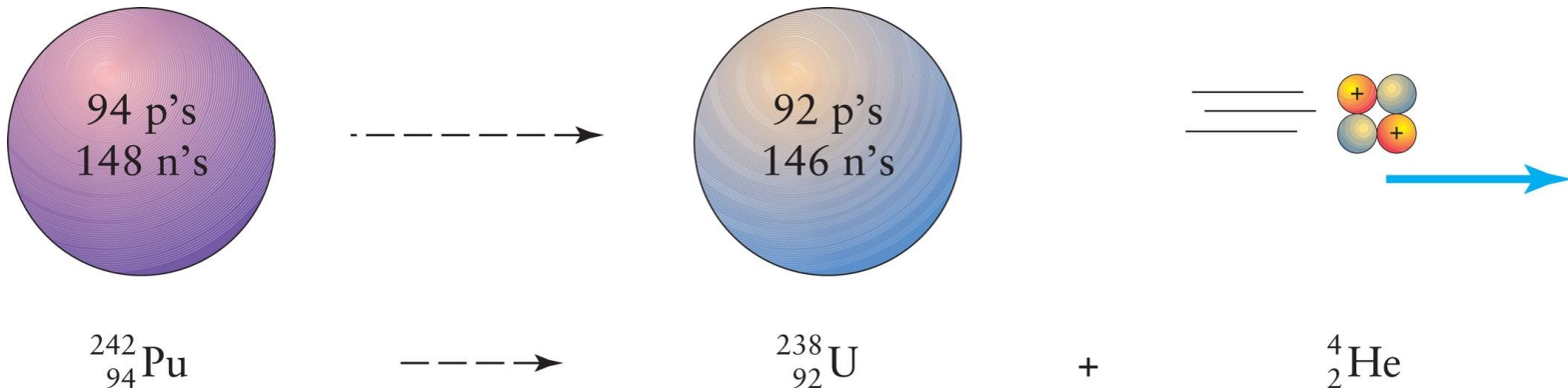


- Beta particles are more penetrating than Alpha ones, however, a sheet of lead can stop a significant fraction of them.

11.2 Beta Decay

Often the daughter nucleus in both alpha and beta decay is itself radioactive and decays into another isotope.

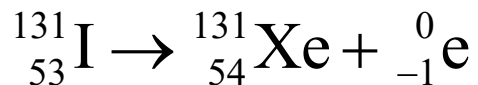
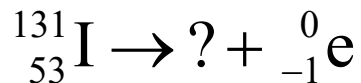
- Plutonium-242 undergoes alpha decay to uranium-238. This is a radioisotope that undergoes alpha decay into thorium-234. This process continues until, after a total of nine alpha decays and six beta decays, the stable isotope lead-206 is reached. This is called a *decay chain*.



Example 11.2

The isotope iodine-131 undergoes beta decay. Write the reaction equation and determine the identity of the daughter nucleus.

From the periodic table, we find that iodine's chemical symbol and atomic number are I and 53, respectively. Therefore, we get the result shown here.

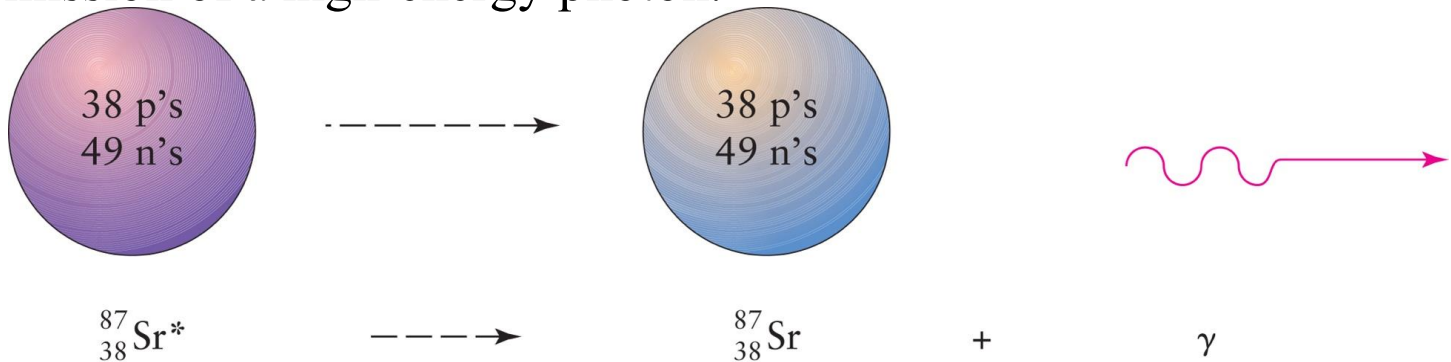


The mass number stays the same, and the atomic number is increased by 1 to 54. From the periodic table, we find that this is the element xenon (Xe).

11.2 Gamma Decay

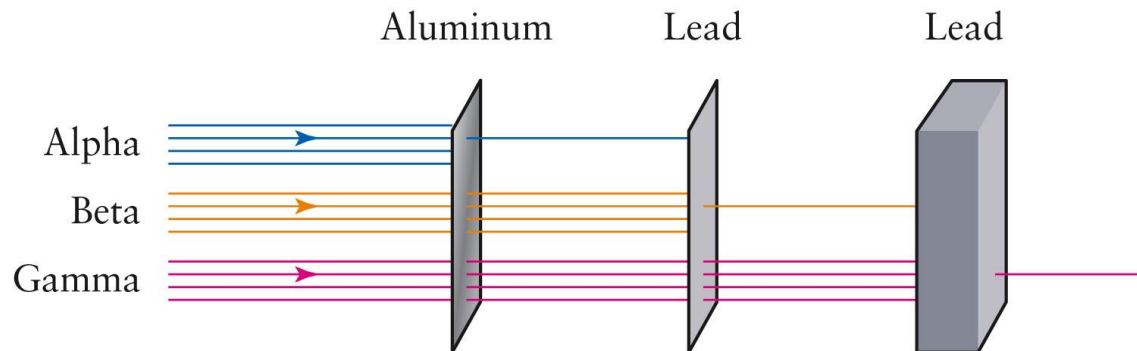
Because a gamma ray has no mass or electric charge, **gamma-ray emission** has *no effect on the mass number A or the atomic number Z of a nucleus*.

- Gamma rays are emitted when the nucleus makes a transition to a lower energy state. The identity of the nucleus is not changed during the process.
- Gamma-ray emission often accompanies alpha decay and beta decay because these two processes often leave the nucleus in an excited state with excess energy that is quickly lost through the emission of a high-energy photon.



11.2 Gamma Decay

- All three types of nuclear radiation are ionizing radiation but differ considerably in their ability to penetrate solid matter.
- **Alpha** particles are the **least penetrating**, they are absorbed after traveling only a short distance.
- **Beta particles are more penetrating** than alpha particles. For comparison, a thin sheet of aluminum that will block essentially all alpha particles will stop only a small fraction of beta particles.
- **Gamma rays are the most penetrating** of the three. Because they have no electric charge and travel at the speed of light. Several centimeters of lead are required to block gamma rays.



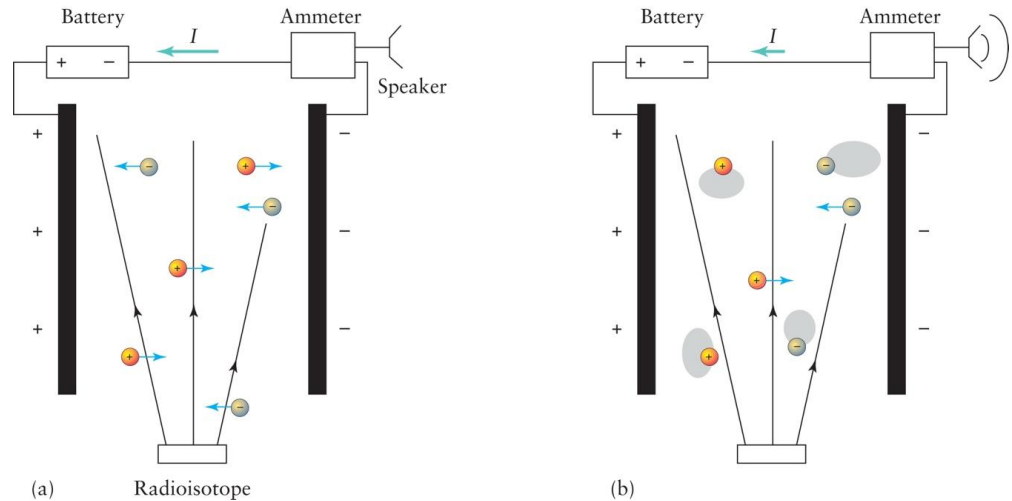
11.2 Applications

Radiation from radioactive decay is used routinely for many purposes:

- Nuclear medicine makes use of radioisotopes for diagnosis and treatment. Dozens of industrial facilities around the world use gamma radiation from cobalt-60 and other isotopes to sterilize disposable medical products.
- The gamma rays can penetrate deep into the interiors of product containers and kill bacteria and other microorganisms that may be present there.

11.2 Applications

Another application of radioactivity is *smoke detector*.



- The alpha radiation ionizes the air between two plates connected to the terminals of a battery.
- Consequently, an electric current flows through the circuit.
- When smoke enters the detection chamber, the ions are attracted to the smoke particles and attach to them. This reduces the flow of charge between the plates.
- An ammeter built into the electronic circuitry detects the resulting decrease in the current and triggers an audible alarm.

11.3 Half-Life

If a nucleus is radioactive, when will it decay?

- In truth, there is no way of knowing precisely when a particular nucleus will decay: it may take a billion years or only a millionth of a second.
- We can predict how much time will elapse, on the average, before a nucleus of a given radioisotope will decay.
- This leads us to the concept of *half-life*.

11.3 Half-Life

DEFINITION

Half-Life: The time it takes for half the nuclei in a sample of a radioisotope to decay. The time interval during which each nucleus has a 50 percent probability of decaying.

11.3 Half-Life

Element	Isotope	Half-Life
Hydrogen	${}^1_1\text{H}$	...
	${}^2_1\text{H}$ (deuterium)	...
	${}^3_1\text{H}$ (tritium)	12.3 yr
Helium	${}^3_2\text{He}$	...
	${}^4_2\text{He}$	...
	${}^5_2\text{He}$	2×10^{-21} yr
	${}^6_2\text{He}$	0.805 s
	${}^8_2\text{He}$	0.119 s
Carbon	${}^{12}_6\text{C}$	...
	${}^{13}_6\text{C}$	...
	${}^{14}_6\text{C}$	5,730 yr
	${}^{15}_6\text{C}$	24 s

Element	Isotope	Half-Life
Silver	${}^{107}_{47}\text{Ag}^*$	44.2 s
	${}^{108}_{47}\text{Ag}$	2.42 min
	${}^{109}_{47}\text{Ag}^*$	39.8 s
	${}^{110}_{47}\text{Ag}$	24.6 s
Uranium	${}^{232}_{92}\text{U}$	70 yr
	${}^{233}_{92}\text{U}$	159,000 yr
	${}^{234}_{92}\text{U}$	245,000 yr
	${}^{235}_{92}\text{U}$	704,000,000 yr
	${}^{236}_{92}\text{U}$	23,400,000 yr
	${}^{237}_{92}\text{U}$	6.75 d
	${}^{238}_{92}\text{U}$	4,470,000,000 yr
	${}^{239}_{92}\text{U}$	23.5 min

Note: Asterisks (*) indicate that the nuclei are in an excited state.

The half-lives of radioisotopes range from a tiny fraction of a second to billions of years.

11.3 Half-Life

- During the span of one half-life, approximately half of the nuclei in a sample will decay—that is, emit their radiation.
- Half of the remaining nuclei will decay during the span of a second half-life, leaving only one-fourth of the original nuclei.
- After three half-lives, one-eighth remain, and so on.
- After ***n half-lives***, ***one-half raised to the power n of the original nuclei will remain undecayed***—as shown in the equation here:

$$N = N_0 \left(1/2\right)^n$$

Here ***N_0 is the original number*** of radioactive nuclei present in the sample, and ***N is the number of radioactive nuclei remaining*** after the elapse of ***n half-lives***.

11.3 Half-Life

For example, let's say that we start with 8 million nuclei of a radioisotope with a half-life of 5 minutes.

$$N = N_0 \left(1/2\right)^n$$

Example:

initially 8 million nuclei

after 5 minutes → 4 million nuclei remain undecayed

after 5 more minutes → 2 million nuclei remain undecayed

after 15 minutes → 1 million nuclei undecayed

after 50 minutes (10 half-lives) →

$$8 \text{ million} \times \left(\frac{1}{2}\right)^{10} = 7,800 \text{ nuclei}$$

Example 11.3

Suppose that 5.5×10^7 atoms of radon are trapped in a basement after which the space is sealed against any further accumulation of the gas. If the half-life of radon is 3.83 days, how many radon atoms remain after one month's time (31 days)?

$$N = N_0 \left(1/2\right)^n$$

$$1 \text{ month} = 31 \text{ days} \times \frac{3.83 \text{ days}}{\text{half-life}} = 8.1 \text{ elapsed half-lives}$$

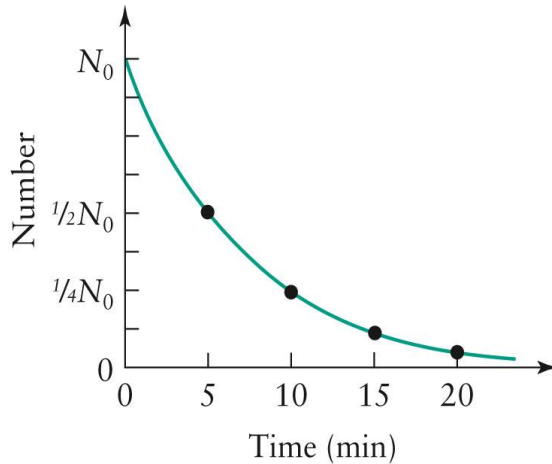
$$\text{initial radon atoms reduced by factor of } \left(\frac{1}{2}\right)^8 = \frac{1}{256}$$

$$\rightarrow \frac{(5.5 \times 10^7 \text{ atoms})}{256} = 2.15 \times 10^5 \text{ atoms}$$

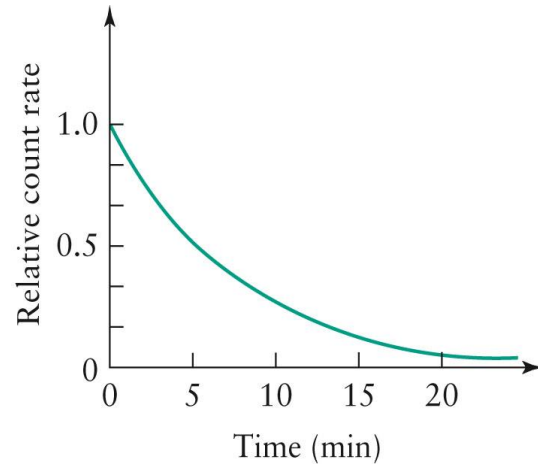
11.3 Half-Life

In reality, it isn't possible to count how many nuclei are left undecayed every 5 minutes.

But with a Geiger counter, you can keep track of how the rate of decay—that is, the number of nuclei that decay each minute—decreases.



(a)



(b)

11.3 Half-Life

Is it useful to know the half-lives of radioisotopes?

Yes, knowing the half-life of a radioisotope allows us to estimate over what period of time a sample will emit appreciable amounts of radiation.

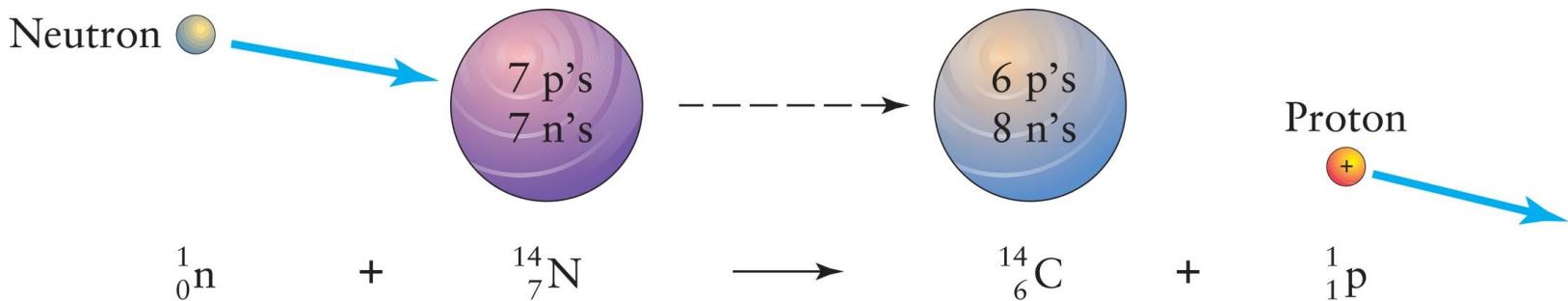
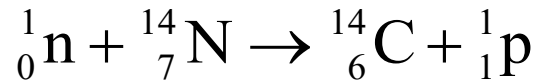
- The isotope americium-241 is used in smoke detectors partly because its long half-life (432 years) ensures that it will continue to produce the ions needed for the detector's operation throughout the device's lifetime.
- Small amounts of certain radioisotopes are sometimes used medically in the human body. The flow of the isotope through the bloodstream can be monitored with a Geiger counter. The radioisotope that is used must have a long enough half-life to remain detectable but not too long, to minimize any possible harmful effects.

11.3 Radioactive Dating

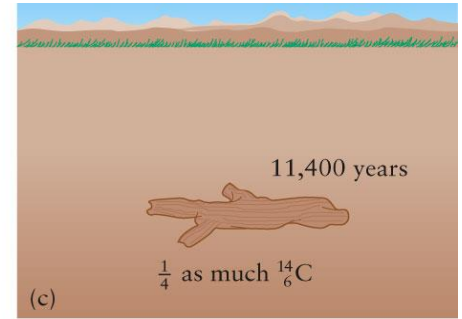
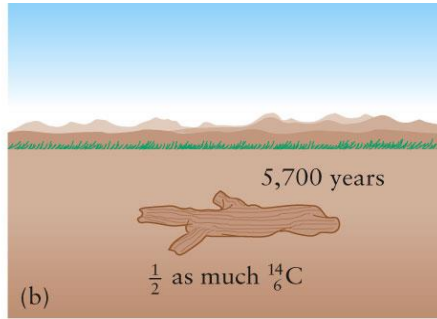
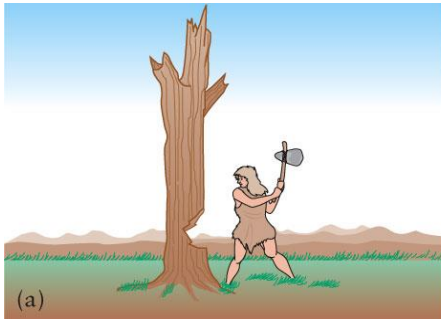
- The regular rate of decay of radioisotopes can be used like a clock.
- Carbon-14 dating is a good example. Carbon is a key element in all living things on Earth.
- Carbon enters the food chain as plants take in carbon dioxide from the air. The complex carbon-based molecules formed by plants are ingested by animals and people.
- About 99 percent of this carbon is the stable isotope carbon-12. About one out of every trillion carbon atoms is radioactive carbon-14.

11.3 Radioactive Dating

- In the upper atmosphere, carbon-14 is constantly being produced by high-energy charged particles from outer space colliding with atoms in air molecules.
- Some of the fragments are free neutrons that cause the formation of carbon-14 when they collide with nitrogen-14. The reaction goes as shown here.



11.3 Radioactive Dating

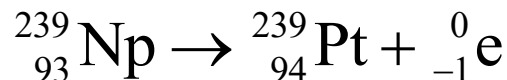
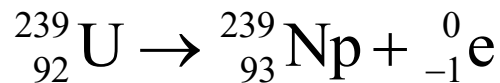
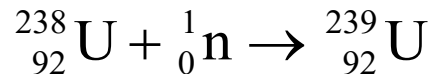


- After an organism dies, no new carbon-14 is added, so the percentage of carbon-14 decreases as the nuclei undergo radioactive decay. This can be used to measure the age of the remains.
- For example, if a tree is cut down and used to build a shelter, 5,700 years later there will be one-half as much carbon-14 in the wood as in a live tree. After 11,400 years, there would be one-fourth as much, and so on.
- Thus, **by measuring the carbon-14 content** in ancient logs, charcoal, bones, fabric, or other such artifacts, **archaeologists can estimate their age.**

11.4 Artificial Nuclear Reactions

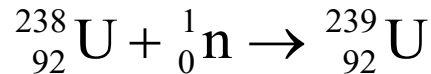
Many other types of nuclear reactions can be induced artificially in laboratories.

- A common example of this is the bombardment of nuclei with alpha particles, beta particles, neutrons, protons, or other nuclei. If a nucleus “**captures**” the bombarding particle, *it will become a different element or isotope*. This is called **artificial transmutation**.
- Note that here something is added to the nucleus, whereas in radioactive decay something leaves the nucleus.

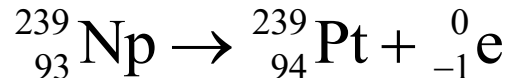
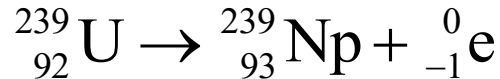


11.4 Artificial Nuclear Reactions

One example of a useful artificial reaction involves bombardment of uranium-238 nuclei with neutrons. The result is uranium-239, as:



This new nucleus undergoes two beta decays, resulting in plutonium-239.



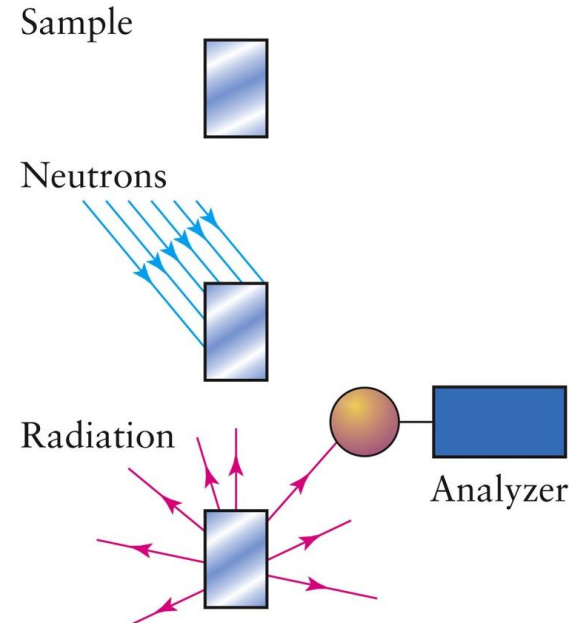
The plutonium-239 can be used directly in nuclear reactors, but the original uranium-238 can't.

- A **breeder reactor** is a nuclear reactor designed to use neutrons to produce, or “breed,” reactor fuel: plutonium-239. This process also shows how elements with Z greater than 92 can be produced artificially.

11.4 Artificial Nuclear Reactions

Neutron activation analysis is an accurate method for determining what elements are present in a substance.

- The material to be tested is bombarded with neutrons. Many of the nuclei in the substance are transformed into radioisotopes.
- By monitoring the intensity and energy of the nuclear radiation that is then emitted, it is possible to determine which elements were originally present.



11.5 Nuclear Binding Energy

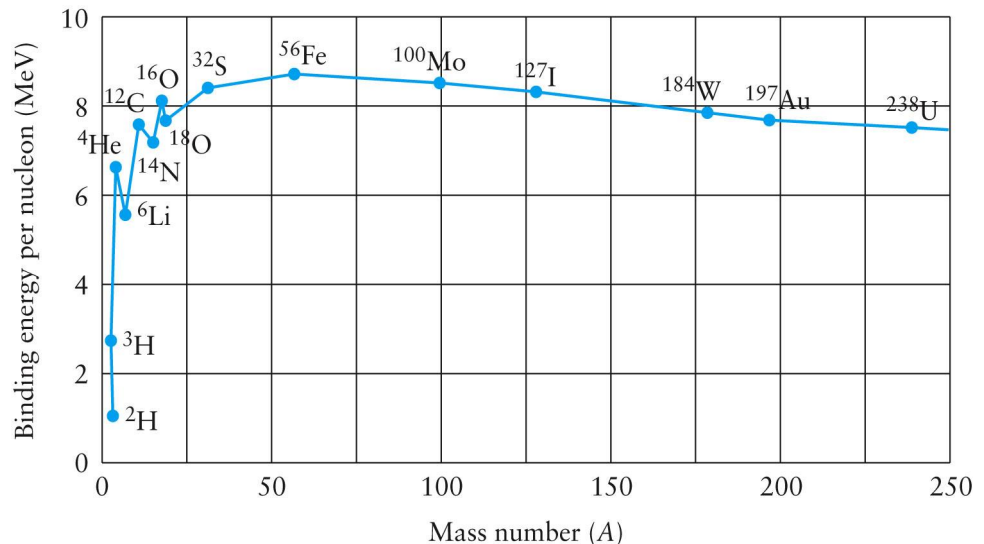
- ❑ If a nucleus is dismantled by removing each proton and neutron one at a time, and the total amount of work done in the process of overcoming the strong nuclear force that binds the nucleons is measured.
- ❑ If the process is now reversed, and these protons and neutrons are allowed to reassemble under their mutual strong force attraction to form the original nucleus, an amount of energy equal to the previous work done would be released.
- ❑ This energy is called the *binding energy of the nucleus*. It indicates how tightly bound a nucleus is.

11.5 Nuclear Binding Energy

- A useful quantity for comparing different nuclei is the average binding energy for each proton and neutron: this is called *the binding energy per nucleon*. It is just the total binding energy divided by the total number of protons and neutrons in the nucleus—that is, the mass number.
- *Nuclei with mass numbers around 50 have the highest binding energies per nucleon: the protons and neutrons are more tightly bound to the nucleus than in larger or smaller nuclei, so these nuclei are the most stable.*

11.5 Nuclear Binding Energy

- The graph of binding energy per nucleon suggests how we can exploit nuclear energy.
- Imagine taking a large nucleus with A around 200 and splitting it into two smaller nuclei. The graph shows that each of these smaller nuclei—say, with A around 100—has a higher binding energy per nucleon than the original nucleus. All of the neutrons and protons have become more tightly bound together and have released energy in the process.



11.5 Nuclear Binding Energy

□ How do we measure the binding energy of a nucleus?

- It is not practical to dismantle a nucleus and keep track of the amount of work done.
- Remember that the largest nuclei contain more than 200 nucleons. The best way to measure binding energy is to use the equivalence of mass and energy.
- The exact relationship between the energy of a particle and its mass is the famous Einstein's equation shown here.

$$E = mc^2$$

- When a particle is given energy, its mass increases; and when it loses energy, its mass decreases. In other words, *mass can be converted into energy and vice versa.*

11.5 Nuclear Binding Energy

For example, the mass of a hydrogen-1 atom (one proton and one neutron):

- The mass of one proton is 1.00785 u,
- The mass of one neutron is 1.00869 u.

The total mass of a hydrogen atom and a free neutron is 2.01654 u.

- But careful measurements of the mass of a hydrogen-2 (deuterium) atom give the value 2.01410 u.

Proton



Neutron



Mass is
0.00244 u less

11.5 Nuclear Binding Energy

- When the neutron and the proton are bound together in the hydrogen-2 nucleus, their combined mass is 0.00244 u less than when they are apart.
- *This much mass “mass defect” is converted into energy when a proton and a neutron combine*, and this energy is the binding energy of the hydrogen-2 nucleus.
- The energy equivalent of this “mass defect” is the total binding energy of the nucleus.

Proton



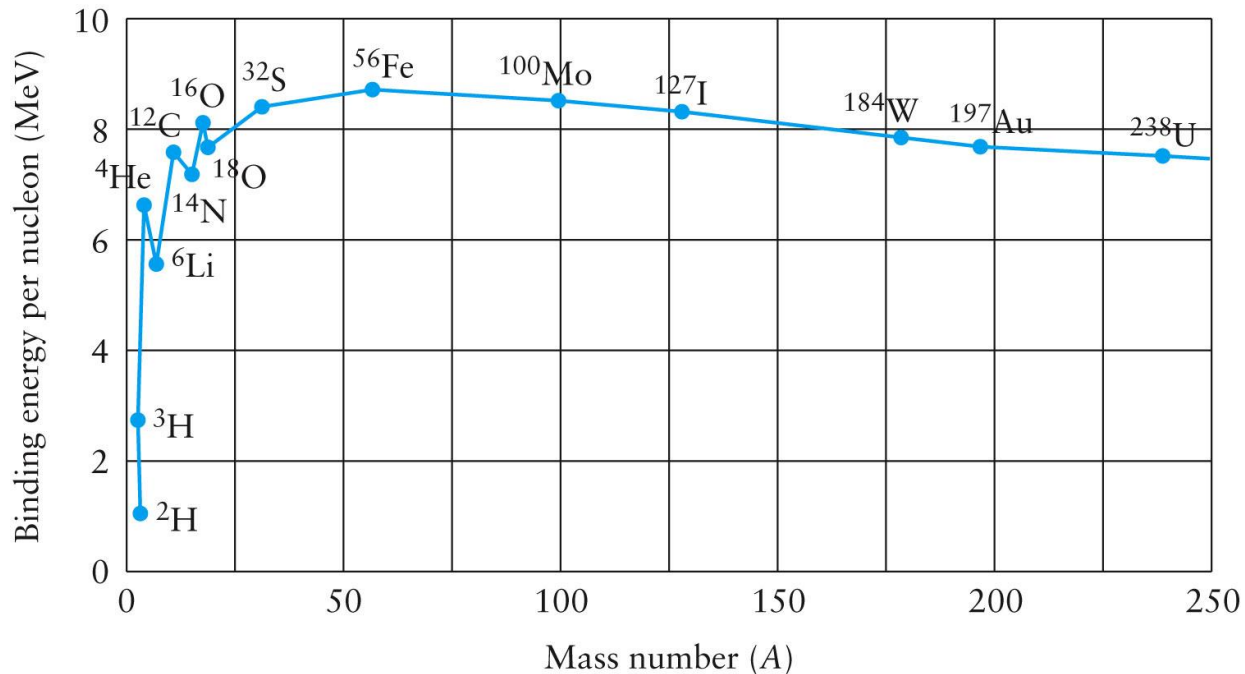
Neutron



Mass is
0.00244 u less

Example 11.4

Compute the binding energy for hydrogen-2 (deuterium) in joules and MeV. Find the binding energy per nucleon for this species and compare the result with that given in the figure.



Example 11.4

$$E = mc^2$$

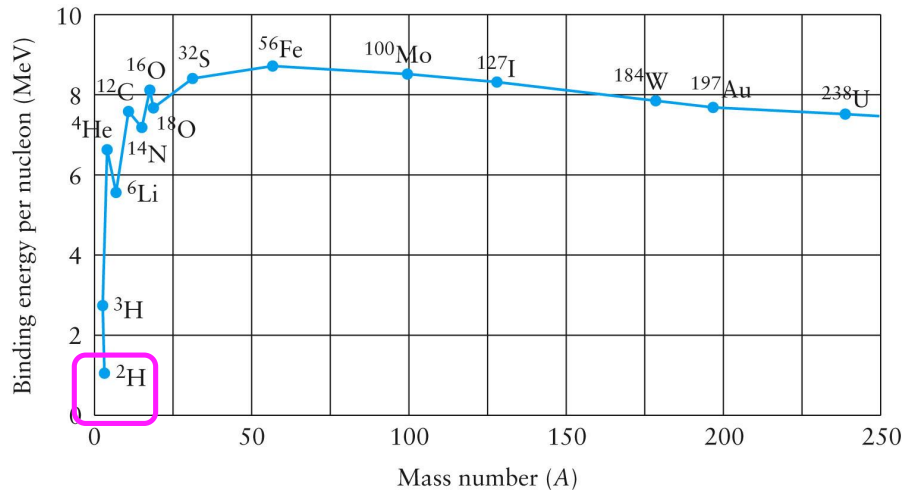
$$= (0.00244 \text{ u} \times 1.66 \times 10^{-27} \text{ kg/u}) c^2$$

$$= 4.0504 \times 10^{-30} \text{ kg} \times (3 \times 10^8 \text{ m/s})^2 = 3.6454 \times 10^{-13} \text{ J}$$

$$\rightarrow (3.6454 \times 10^{-13} \text{ J}) / (1.602 \times 10^{-19} \text{ J/eV}) = 2.275 \text{ MeV}$$

The hydrogen-2 nucleus contains two nucleons (one proton and one neutron). The binding energy per nucleon for deuterium is thus

$$\begin{aligned} & 2.275 \text{ MeV} \div 2 \\ & = 1.1375 \text{ MeV} \end{aligned}$$



11.6 Nuclear Fission

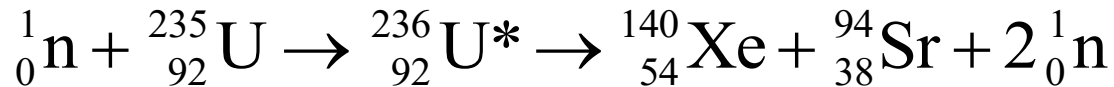
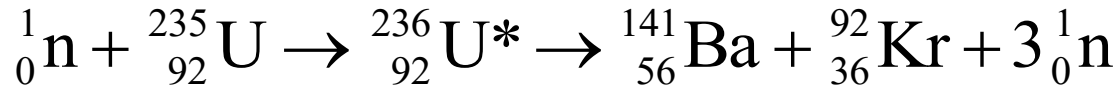
DEFINITION

Nuclear Fission: The splitting of a large nucleus into two smaller nuclei. Free neutrons and energy are also released.

- The two resulting nuclei are called *fission fragments*.
- Fission is exploited in nuclear power plants and atomic bombs

11.6 Nuclear Fission

- Fission is commonly induced by bombarding nuclei with neutrons.
- After absorbing a neutron, the nucleus becomes highly unstable and quickly divides.
- Two of the many possible fission reactions of uranium-235 are shown here.



- The asterisk (*) indicates that the uranium-236 is in an excited, unstable state.
- The total mass of the fission fragments and the neutrons is less than the mass of the original uranium nucleus and the neutron. The missing mass is converted into energy.

11.6 Nuclear Fission

In addition to the energy released during fission, two other aspects of this process are extremely important.:

1. Most fission fragments are radioactive, and they undergo beta decay.
2. The neutrons released during fission can strike other nuclei and cause them to split, causing a *chain reaction*.

11.6 Nuclear Fission

- ❑ If just one of the neutrons from each fission induces another fission, a *stable chain reaction results*. The number of nuclei that split each second remains constant.
 - *Energy is therefore released at a steady rate.*
 - This type of reaction is used in **nuclear power plants**.

- ❑ If, on the other hand, more than one of the neutrons causes other fissions, an *unstable chain reaction results*—a nuclear explosion. One fission could trigger two, these two could trigger four more, then eight, sixteen, and so on.
 - *Energy is released at a rapidly increasing rate.*
 - This is the process used in **atomic bombs**.

11.6 Atomic Bombs

To produce an *uncontrolled fission chain reaction* for the atomic bomb:

- we must ensure that the fission of each nucleus leads to more than one additional fission.
 - This is accomplished by using a *high density of fissionable nuclei* so that each neutron emitted in a fission is likely to encounter another nucleus and cause it to fission.
- Also, there must be a sufficient number of fissionable nuclei (uranium-235 or plutonium-239) and they must be put into the proper configuration—a sphere, for example.
 - This increases the probability that each neutron will collide with a nucleus before it escapes.

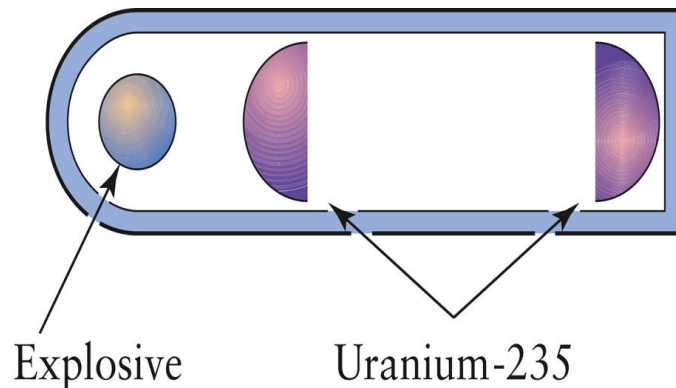
11.6 Atomic Bombs

When these conditions are met, there is a *critical mass of fissionable material*.

- If there are too few fissionable nuclei present or if they are spread out too far, too many of the neutrons will escape without causing other fissions.
- Also, the critical mass must be held together long enough for the chain reaction to cause an explosion.
- A premature explosion is highly undesirable. This is prevented by keeping the fissionable material out of the critical mass configuration before the explosion. Two different techniques are used: a gun-barrel atomic bomb and the implosion bomb.

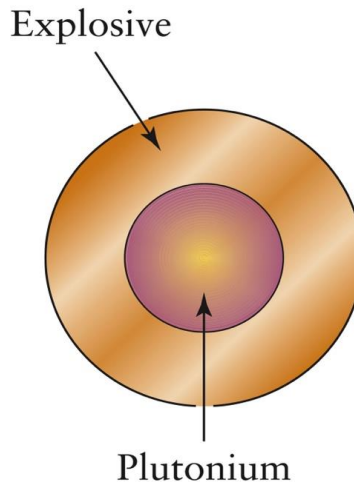
11.6 Atomic Bombs

- In a gun-barrel atomic bomb, two subcritical lumps of uranium-235 are placed at opposite ends of a large tube.
- The explosion is triggered by forcing the two lumps together into a critical mass.
- This type of bomb was dropped on Hiroshima, Japan.



11.6 Atomic Bombs

- In the implosion bomb, plutonium-239 is used . A subcritical sphere of the plutonium is surrounded by a shell of specially shaped conventional explosives.
- These explosives squeeze the plutonium-239 so much so that it becomes a critical mass, and an explosion occurs.
- This type of bomb was dropped on Nagasaki, Japan.



11.6 Nuclear Power Plants

- About one-eighth of the world's electricity is generated by nuclear power plants.
- The reactors in these plants release the energy from nuclear fission in a *controlled manner*.
- The key to preventing a fission chain reaction from escalating to an explosion is *controlling how the neutrons induce other fissions*.



11.6 Nuclear Power Plants

To controlling how the neutrons induce other fissions. There are different ways to do so, such as:

1) Using only *slightly enriched uranium fuel*, typically about 3 percent uranium-235.

- This keeps the density of fissionable nuclei low so that not all neutrons from each fission are likely to strike other U-235 nuclei and induce other fissions, and it makes it impossible for a critical mass configuration to arise: a nuclear reactor cannot explode like an atomic bomb.

11.6 Nuclear Power Plants

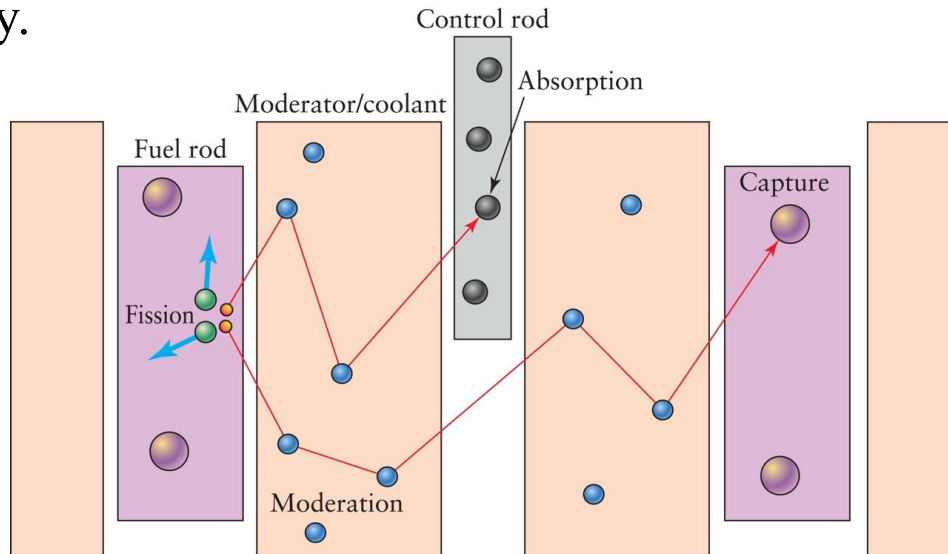
2) Another important factor in producing and controlling a fission chain reaction in a nuclear reactor is the fact that *slow neutrons* are much better at causing uranium-235 nuclei to undergo fission than fast neutrons.

- How are the neutrons slowed down?
- This is done through the use of a moderator (water and graphite), a substance that contains a large number of small nuclei. As neutrons pass through the moderator, they collide with the nuclei and lose some of their energy: they slow down.

11.6 Nuclear Power Plants

3) The uranium fuel is shaped into long rods that are kept separate from each other.

- Each rod relies on thermal neutrons from neighboring rods to sustain the chain reaction.
- This makes it possible to control the chain reaction by lowering control rods between the fuel rods.
- *Control rods* are made of materials that absorb neutrons effectively.



11.7 Nuclear Fusion

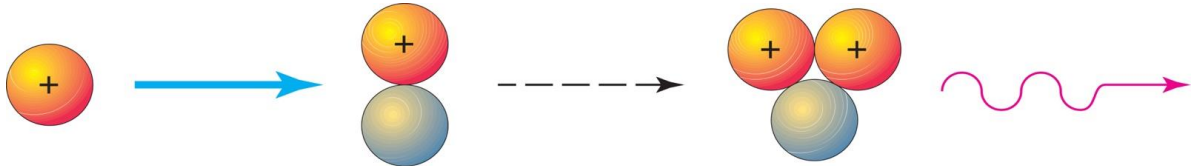
DEFINITION

Nuclear Fusion: The combining of two small nuclei to form a larger nucleus with the accompanying release of energy.

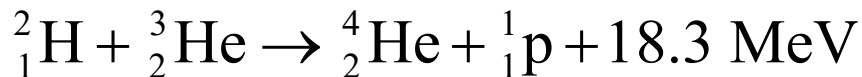
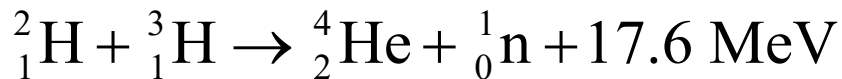
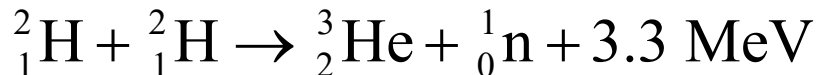
- ❑ Energy is released because the total mass of the nucleons after the fusion is less than the total mass before. The missing mass is converted into energy.
- ❑ Fusion is the source of energy in the Sun, the stars, and hydrogen bombs.

11.7 Nuclear Fusion

- One example of a fusion reaction is shown in the figure.
- In this case, two hydrogen nuclei, one hydrogen-1 and the other hydrogen-2 (deuterium), fuse to form a nucleus of helium-3.

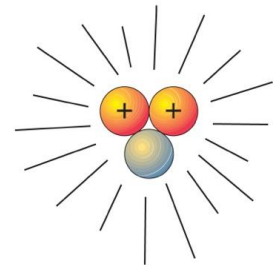
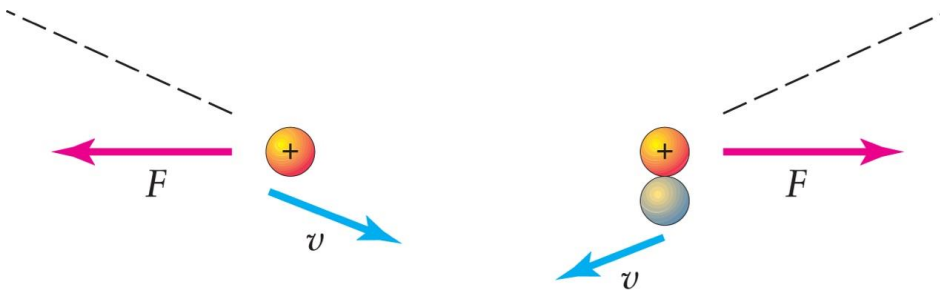


- There are many other possible fusion reactions that result in a release of energy, such as:



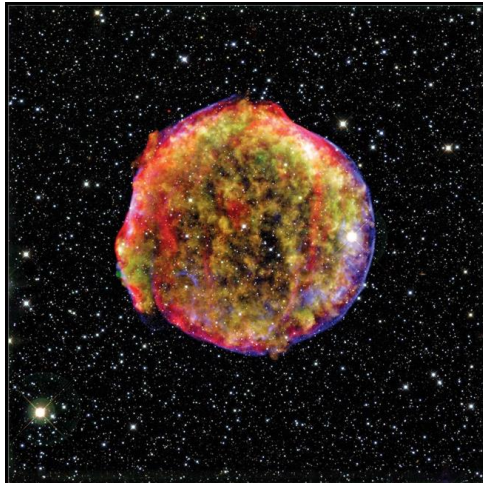
11.7 Nuclear Fusion

- Fusion can occur only when the two nuclei are close enough for the short-range nuclear force to pull them together.
- However, nuclei are positively charged. When brought close together, they exert strong repulsive forces on each other.
- How can this difficulty be overcome?
- The most common way is with *extremely high temperatures*.
- If nuclei can be given enough average kinetic energy, their inertia will carry them close enough together to fuse when they collide. This process is called *thermonuclear fusion*.



11.7 Fusion in Stars

- ❑ The Sun is composed mostly of hydrogen in a dense, high-temperature plasma.
 - At the Sun's core, the temperature is around 15 million degrees Celsius, and the pressure is more than 1 billion atmospheres.
 - Under these conditions, the hydrogen undergoes a series of fusion reactions that results in the formation of helium.



11.7 Hydrogen Bombs

- ❑ Hydrogen bombs get most of their explosive energy from the fusion of hydrogen.
 - To produce the high temperatures necessary for the fusion reactions to occur, a nuclear fission explosion is used as a trigger.
 - The fission explosion, an incredibly huge blast in itself, is a primer for the monstrous fusion explosion.
 - In terms of the relative amount of energy released, a hydrogen bomb is to an atomic bomb what a stick of dynamite is to a small firecracker.

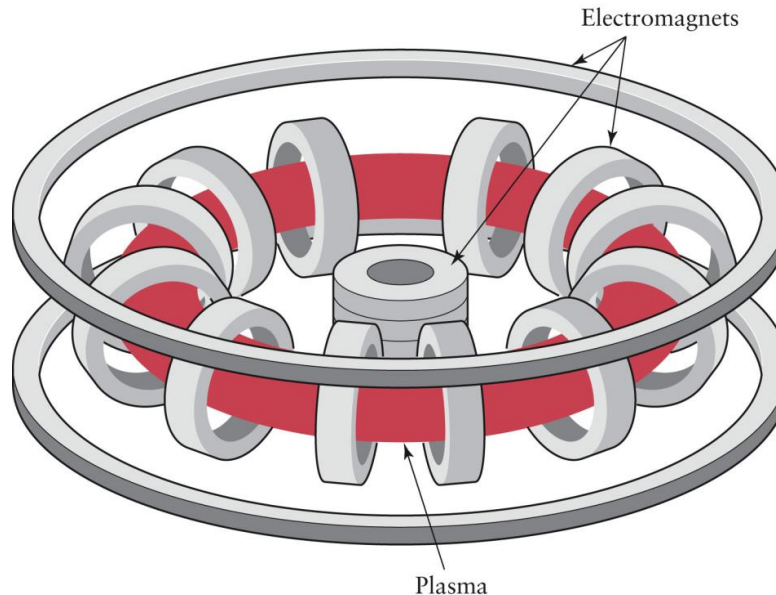
11.7 Controlled Fusion

For a controlled fusion to occur, two conditions must be met:

- 1) The nuclei must be raised to an extremely high temperature to ignite the fusion reaction
 - It is possible to produce temperatures in the millions of degrees, but the problem is containing a plasma at this high temperature. Containers made of conventional materials would melt long before such temperatures would be reached.
- 2) High density of nuclei for the probability of collisions and, therefore, fusions, to remain high.

11.7 Controlled Fusion

- Specially shaped magnetic fields are used to keep the plasma confined without letting it come into contact with other matter.
- This is possible because the nuclei are charged particles and consequently experience a force when moving in a magnetic field.
- One of the most promising designs, called a *tokamak*.



Important Equations

IMPORTANT EQUATIONS

Equation	Comments
$N = N_0 (1/2)^n$	Radioactive decay
$E = mc^2$	Einstein's mass-energy relation